

<u>STUDENT ID:</u>	<u>FULLNAME:</u>	<u>SIGNATURE:</u>
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Q.1) In the context of data science projects, the data understanding phase is crucial for ensuring the success of subsequent steps. Describe in detail the critical steps involved in the data understanding phase. Your answer should include the purpose of each step, the methods and techniques used, and the key outcomes expected.

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Q.2) Identify two examples of supervised learning algorithms from the list below. For each selected algorithm, provide a brief explanation of how it works.

- A) Decision Trees
- B) K-Means Clustering
- C) Support Vector Machines (SVM)
- D) Apriori Algorithm

First algorithm:

Second algorithm:

Q.3) Identify two methods from the list below that can be used for anomaly detection. For each selected method, provide a brief explanation of how it works.

- A) Isolation Forest
- B) Autoencoders
- C) Gradient Boosting
- D) Decision Tree Classifier

First method:

Second method:

Q.4) Principal Component Analysis (PCA) is a popular technique used in data science for dimensionality reduction.

a) Explain the main objectives of PCA.

b) Explain how PCA helps in identifying the most significant features.

c) Describe the steps involved in performing PCA on a dataset.

